

MONK'S HOOD CACTUS

Astrophytum ornatum

The genus name for this cactus, *Astrophytum*, means “star plant” in ancient Greek. It was first collected in 1827 by Thomas Coulter, an Irish doctor and botanist, who sent it to Professor de Candolle at the Botanic Garden of Geneva. When de Candolle unpacked the cactus he thought it was covered in fungus, but discovered that the white spots were actually tufts of hair or trichomes. These woolly scales may help the cactus collect water and protect it against the sun, but they also provide camouflage. This species is the most densely covered of all the *Astrophytum* cacti, and also the spiniest. It is now rare in the wild.

SIZE 1.5 m (5 ft)
HABITAT Semi-desert scrub
DISTRIBUTION N.E. Mexico
LEAF TYPE Spines

< ROOT SYSTEM

Fibrous and shallow roots help the cactus to absorb water from a large area, essential in an environment where after brief showers only a few centimetres of soil are wet.

OUTER PETALS >

The numerous, narrow outer petals of the flower are pale yellow and have brown tips. The flower itself grows to 11 cm (4½ in) in diameter.

< WOOLLY SCALES

This cactus has white, scale-like tufts of trichomes arranged in bands between its ribs. They are densely woolly in young plants but become sparse with age.

spine protects the slow-growing plant from being eaten

pith, used for water storage

fibrous roots

petal

stamen (male part that produces and releases pollen)

stigma

style (female part that connects the stigma to the ovary)

ovary

> FLOWER SECTION

The yellow petals have a broad oblong shape with a slightly serrated apex. The flower has a yellow carpel (the female reproductive organs, made up of the stigma, style, and ovary) and yellow stamens.

^ OVARY

Located below all the other reproductive parts of the flower, the ovary contains ovules that develop into seeds.

^ STIGMA

The flower has a single stigma with 7–12 lobes. This is the part of the flower used to catch pollen. The stigma is about 1.5 cm (½ in) long.